Course Name: Computer Networks

Course Outcomes

CO1: Understand the concept of OSI and TCP/IP reference model
CO2: Understand the basics of data transmission at physical layer.
CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
CO4: Apply error detection and correction technique to eliminate transmission error.
CO5: Analyze the fixed and variable length address (IPv4) subneting for the given scenarios.
CO6: Understand the design issues of the transport layer.
CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8: Understand IPv6 addressing and differentiate it from IPv4.

Printed Pages: 02 **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23

B.Tech (CSE), 2nd Year, IV Semester

BCSC0008 Computer Networks

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Name four basic topologies used in computer networks. State the advantages and disadvantage of using STAR topology. Assume 6 devices are arranged in a mesh topology. How many cables are needed? How many ports are needed for each device? OR How do guided media differ from un-guided media? Name the advantages of optical fiber over co-axial cable.	3	2	U	F
2	How does NRZ-L differ from NRZ-I? Sketch the Manchester and Differential Manchester encoding for the bit steam 0001110101.	3	2	A	P
3	Why the flow control is required at Data Link Layer? In what way Selective-Repeat is better than Go-Back-N ARQ technique. OR Explain the working of Virtual Circuit Switching in detail. How it is different from Datagram switching?	3	3	U	C
4	In CRC checksum method, assume that given frame for transmission is 1101011011 and the generator polynomial is $G(x)=x^4+x+1$. After implementing encoder. What will be the encoded word sent from sender side?	3	4	A	P

5	What do you mean by vulnerable time in slotted ALOHA? A slotted ALOHA network transmits 200 bits' frames using a shared channel with 200 kbps bandwidth. If the system (all stations put together) produces 1000 frames per second, then compute the throughput of the system.	3	3	A	P

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	How are OSI and ISO related to each other? Discuss the responsibilities of Physical, Datalink and Network layer.	5	1	A	F
7	Mentioned the condition to detect collision in CSMA/CD. Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km (kilometer) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?	5	3	An	C
8	A sender needs to send the four data items 0x3456, 0xABCC, 0x02BC, and 0xEEEE. Answer the following: a. Find the checksum at the sender site. b. Find the checksum at the receiver site if there is no error. c. Find the checksum at the receiver site if the second data item is changed to 0xABCE.	5	4	A	P

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive). However, For Engg. Courses in addition to F, C, P & M include D-Fundamental Design Principles, S-Criteria and Specifications, PC-Practical Constraints, DI- Design Instrumentalities

Course Name: **ENGINEERING MATHEMATICS III**

Course Outcomes:

CO1- Compute the coefficient of correlation and regression between two variables
CO2- Apply Chi-square test and probability distributions
CO3- Use Laplace transform in solving differential equations
CO4- Know the concept of analytic functions and its applications in engineering
CO5- Understand bilinear transformation and its applications
CO6- Use Cauchy's residue theorem in solving real integrals

Printed Pages: 03 **University Roll No.** .....

Mid Term Examination, Even Semester 2022-23
B.Tech. [CSE (Sec. A to E, Q to X & ME, SM, MTE) II Year, IV Semester
Sub. Code & Name: BMAS 1103, ENGINEERING MATHEMATICS III

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
*Attempt **Both** Sections.*

Section – A

*Attempt **All** Questions.* 3 x 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Using the method of least squares, fit a straight line from the following data: x: 0, 2, 4, 5, 6 y: 5, 10, 15, 21, 30	3	1	A	P
2	The regression equations calculated from a given set of observations for two random variables are: $x = -0.4y + 6.4,$ and, $y = -0.6x + 4.6.$ Calculate: (i) Mean values of x and y (ii) Coefficient of correlation between x and y.	3	1	An	C
3	Calculate the Karl Pearson's coefficient of correlation between the marks obtained by 8 students in Mathematics and Statistics: Maths: 25, 30, 32, 35, 37, 40, 42, 45 Statistics: 8, 10, 15, 17, 20, 23, 24, 25	3	1	U	F

4	Obtain the Laplace transform of following functions: (i) $F(t) = e^{-3t}(2\cos 5t - 3\sin 5t)$ (ii) $F(t) = \begin{cases} (t-1)^2, t > 1 \\ 0, \ t < 1 \end{cases}$ (iii) $F(3t)$ if $L[F(t)] = \frac{1}{p} e^{-\frac{1}{p}}$ Identify and also name the properties of Laplace transform which are applied in getting Laplace transform of the above functions.	3	3	A	P
5	The probability that a bulb produced by a factory will fuse after a use of 150 days is 0.05. Find the probability that out of 5 such bulbs, (i) none, (ii) at most one, (iii) at least one will fuse after 150 days of use. **OR,** Discuss the mean and variance of the Poisson distribution. How is Poisson distribution different from the Binomial distribution? Using Poisson distribution, find the probability that the ace of spades will be drawn from a pack of well-shuffled cards at least twice in 104 consecutive trials.	3	2	E	C

Section – B

Attempt All Questions. 5 x 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	In an experiment on the immunization of goats from anthrox, the following results were obtained. Derive your inferences on the efficiency of the vaccine. It is given that the tabulated value of χ^2 at 5% level of significance for 1 degree of freedom is 3.841.	5	2	A	M

	Died	Survived
Inoculated	2	10
Not inoculated	6	6

7	What is Laplace transform of a function? Can you give an example of a function whose Laplace transform does not exist? Determine the Laplace transform of $F(t) = \frac{e^{-3t} - e^{-6t}}{t}$. Hence or otherwise evaluate: $\int_0^\infty F(t)\,dt$	5	3	An	C
8	A genetical law says that the children having one parent of blood group M and the other parent of blood group N will always be one of the three blood groups M, MN, N and that the average number of children in these groups will be in the ratio 1: 2: 1. The report on an experiment states as follows: **"Of 162 children having one M parent and one N parent, 28.4 % were found to be of group M, 42 % were found to be of group MN and the rest of the group N."** Do these data in the report conform to the expected genetic ratio 1: 2: 1? [It is given that the tabulated value of χ^2 at 5% level of significance for 2 degree of freedom is 5.991.] **OR,** If $F(t)$ is a periodic function with period T then, prove that: $L\{F(t)\} = \frac{1}{1-e^{-pT}} \int_0^T e^{-pt} F(t)\,dt$ Hence or otherwise, find the Laplace transform of the following periodic function: $F(t) = \begin{matrix} 1, & 0 \le t \le 1, 2 \le t \le 3, 4 \le t \le 5, \ldots \\ 0, & 1 \le t \le 2, 3 \le t \le 4, 5 \le t \le 6, \ldots \end{matrix}$	5	2, 3	C	M

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive).

Course Name: Data Structure and Algorithms

Course Outcome

CO1-Understand the concepts of data structure and algorithms.
CO2-Implement of stack using array linked list and its applications
CO3-Implement of queue using array linked list and its applications
CO4- Implement of linked list and its applications
CO5-Implement hierarchical based problems using tree traversal algorithm.

Printed Pages: 2 **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23

B.Tech (CSE/CCV/DA/IIOT/AIML/CSF), 2ND Year, 4th Semester

BCSC 0006/Data Structures and Algorithms

Time: 2 Hours **Maximum Marks: 30**

Instruction for Students:

1. Read question carefully.
2. Don't Overwrite
3. Write the complete code in one place neatly
4. Maintain appropriate indentation while writing code in java if needed.
5. Commenting code is optional.

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	a) Define algorithm. What are the various properties of algorithm? b) Differentiate between static and dynamic data structures. Give suitable examples.	1.5 1.5	CO 1	U	C
2	Write a class that includes methods to a) Count the total number of nodes in the linked list b) Insert the node at beginning in the doubly linked list.	3	CO 4	AN	F
3	Define recursion? Differentiate between recursive statement and iterative statement. Write the algorithm of recursive function to print the Fibonacci series.	3	CO 2	C	C
4	Define stack. Write the code to implement stack using linked list.	3	CO 2	R	P
5	a) The following sequence of operations is performed on a stack having maximum size=8. Stack: P, Q, R, S, __, __, __, __ PUSH(K), PUSH(L), POP (), POP (), POP (), PUSH(M), PUSH(N), PUSH(O), PUSH(T). b) What do you mean by worst case, average case and best case complexity? Give suitable example for each.	1.5 1.5	CO 2	E,A N	C

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	What do you mean by queue? Explain all the types of queue with diagram. Which queue you prefer to use and also give the suitable reason. Write the methods to insert from front and deletion from rear in the double ended queue.	5	CO 3	R	P
7	a) What do you mean by complexity? Explain two types of complexity. Also discuss in detail all types of asymptotic notation with diagram. b) Study the following infix expression: Convert this expression into a prefix and show the stack table to evaluate it. ((A + B) * D) ^ (E-F)	2.5 2.5	CO 2	C, E	P
8	What do you mean by data structure? Discuss in detail various operations of data structures. Also explain all the classification of data structure with suitable example.	5	CO 1	U	M

Course Name: Introduction to Microprocessor

Course Outcome

CO1: Demonstrate the Microprocessor internal architecture and its operations.

CO2: Develop programs based on 8085 microprocessor instruction set and addressing mode.

CO3: Develop program using looping, counting, indexing, counter and time delays.

CO4: Understand the concept of stack and subroutine for modular approach.

CO5: Compare accepted standards and guidelines to select microprocessor (8085 & 8086) to meet performance requirements.

CO6: Analyze the concept of interfacing the processor to external device with I/O programming & Interrupt Driven I/O.

CO7: Understand the working of interfacing chips (8237, 8253/54, 8255 & 8259)

Printed Pages: **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23

B.Tech (CSE/AIML/DA/IIoT/CCV/CSF), II-Year, IV-Semester

BCSC 0007: Introduction to Microprocessors

Time: 2 Hours **Maximum Marks: 30**

Attempt All Questions.

Section – A **3 X 5 = 15 Marks**

No.	Detail of Question	Marks	CO	BL	KL
1	Draw and explain the flag register of 8085 microprocessor.	3	1	U	C
2	Specify the contents of accumulator & the status of carry flag after the execution of each instruction: **MVI A, 80H** **ORA A** **RAR**	3	2	U,R	F.C
3	Explain the function of the following signals: a) *READY* b) *IO/M'* c) *RESET OUT*	3	2	U	C
4	Write an assembly language program to Unpack the BCD number stored in memory location	3	3	U	F.C

	20(0H and store the two digits in memory loc itions 2100H and 2101H such that memory loc ition 2100H will have lo\ er BCD digit.				
5	Ide itify the number of mac iine cycles and T-Sta es for the execution of the following ins ructions: (a) ADD B (b) LHLD, 2500H (c) CPI, 85H	3	2	Ap	C.P

Section – B

5 X 3 = 15 Marks

No.	Detail of Que tion	Marks	CO	BL	KL
1	With the help of neat di gram, explain the arc itecture of 8085 microp ocessor in detail.	5	1	U	C
2	Wr te the function and addressing mode of the fol owing instructions: (d) XCHG (e) DAD B (f) CPI, 85H (g) POP PSW (h) SIM	5	2	Ap	C.P
3	Gi e the sequence of machine cycles during the execution of the ***OUT, 50H*** instruction by the 8085 processor. Also, draw its timing diagram.	5	3	Ap	C,P

Course Name: DISCRETE MATHEMATICS

Course Outcome

CO1: Understand the notion of mathematical thinking and proofs to solve the problem.
CO2: Apply the basics of discrete probability and number theory to solve the real world problem.
CO3: Analyze basic discrete structures and algorithms using effectively algebraic techniques.
CO4: Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
CO5: Analyze the validity of an argument using logical notation.
CO6: Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
CO7: Understand the basic principles of sets, set equalities and operations in sets.
CO8: Apply counting principles to determine probabilities.

Printed Pages:01

University Roll No. .....

Mid Term Examination, Even Semester 2022-23
B.Tech. (CSE), II Year, IV Semester
Subject Code: BCSC1010, Subject Name-Discrete Mathematics

Time: 2 Hours

Maximum Marks: 30

Instruction for students:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section – A

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Show that for any two sets A and B in set theory: $A - (A \cap B) = A - B$. If $A = \{2,4,6,8\}$ *and* $B = \{3,5,6,7\}$ then find A – B ?	1+2	CO1, CO7	U	P
2	In a renowned software development company of 240 computer programmers 102 employees are proficient in Java, 86 in C#, 126 in Python, 41 in C# and Java, 37 in Java and Python, 23 in C# and Python, and just 10 programmers are proficient in all three languages. How many computer programmers are there those are not proficient in any of these three languages?	3	CO2	An	F
3	Use mathematical induction to prove that $n^3 - n$ is divisible by 3 whenever n is a positive integer.	3	CO1	A	C
4	A bag contains 10 red marbles, 10 white marbles, and 10 blue marbles. What is the minimum no. of marbles you have to choose randomly from the bag to ensure that we get 4 marbles of same color?	3	CO1, CO5	R, E	M

5	Find the solution to the recurrence relation $a_n - 6a_{n-1} + 11a_{n-2} - 6a_{n-3} = 0$ given $a_0=20$, $a_1=5$ and $a_2=15$	3	CO2	E	P

Section – B

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is the Cartesian product of A = {1, 2} and B = {a, b, c}? Show that the relation R is an equivalence relation in the set A = {1, 2, 3, 4, 5} given by the relation R = {(a, b) :|a-b| is even}.	1+4	CO1	A, An	C
2	Solve the recurrence relation $a_n = 3a_{n-1} - 2a_{n-2}$ by the generating function method with initial conditions $a_0=1$ and $a_1=3$	5	CO2	E	P
3	i) How many numbers are there between 99 and 1000, having at least one of their digits 7? ii) Find the number of permutations of the letters of the word ALLAHABAD.	2+3	CO8	U, E	C, M

Printed Pages: 13 University Roll No.....

Mid-Term Examination, Even Semester 2022-23

B.Tech. CSE, II/III: Year, IV/VI: Semester

BCSE 0903: Data Science and Analytics

Time: 1-Hour **Maximum Marks: 30**

Note: Attempt all questions. **60 x 0.5 = 30 Marks**

1. Which of the following is the basic data structure in pandas?
   a. Series
   b. Scaler
   c. Vector
   d. Matrix

2. How do you read a CSV file in pandas?
   a. pandas.read_csv()
   b. pandas.load_csv()
   c. pandas.import_csv()
   d. pandas.fetch_csv()

3. What is Pandas?
   a. A programming language
   b. A data analysis library in Python
   c. A machine learning algorithm
   d. A statistical software

4. Which of the following operations can be performed on a Pandas Series?
   a. Filtering

b. Arithmetic operations
c. Sorting
d. All of the above

5. How can you select a single column in a DataFrame?
   a. df['column_name']
   b. df[column_index]
   c. df.select_column('column_name')
   d. df.column_name

6. How can you select multiple columns in a DataFrame?
   a. df['column1', 'column2']
   b. df[['column1', 'column2']]
   c. df[column1, column2]
   d. df.column1, column2

7. How can you drop rows with missing values in a DataFrame?
   a. df.dropna()
   b. df.fillna()
   c. df.replace()
   d. df.interp()

8. How can you drop a column in a DataFrame?
   a. df.remove('column_name')
   b. df.drop(column_index)
   c. df.drop('column_name')
   d. df.delete(column_index)

9. How can you concatenate two or more DataFrames?
   a. df.concat([df1, df2])
   b. df.append([df1, df2])
   c. df.merge([df1, df2])
   d. df.combine([df1, df2])

10. How can you remove duplicate rows from a DataFrame?
   a. df.unique()
   b. df.remove_duplicates()
   c. df.filter_duplicates()
   d. df.drop_duplicates()

11. How can you set a new index in a DataFrame?
   a. df.set_index('column_name')
   b. df.column_name.set_index()
   c. df.index('column_name')
   d. df.index = 'column_name'

12. What is the method to check for missing values in a pandas DataFrame?
   a. notnull()
   b. isnull()
   c. isna()
   d. notna()

13. What is the function to merge two dataframes in pandas?
   a. concat()
   b. join()
   c. merge()
   d. unite()

14. What is the function to describe the summary statistics of a pandas DataFrame?
   a. summary()
   b. describe()
   c. analyze()
   d. stat()

15. What is the method to fill missing values in a pandas DataFrame or Series?

a. update()
b. replace()
c. fillna()
d. impute()

16. What is the method to find unique values in a pandas DataFrame or Series?
a. unique()
b. nunique()
c. count_unique()
d. get_unique()

17. How can you get the top n rows of a DataFrame?
a. df.top(n)
b. df.first(n)
c. df.head(n)
d. df.select(n)

18. How can you get the bottom n rows of a DataFrame?
a. df.tail(n)
b. df.last(n)
c. df.bottom(n)
d. df.select(n, from_end=True)

19. How can you rename columns in a DataFrame?
a. df.column_name.rename('new_name')
b. df.columns.rename('old_name', 'new_name')
c. df.columns = {'old_name': 'new_name'}
d. df.rename(columns={'old_name': 'new_name'})

20. Which of the following is not a type of data analytics?
a. Descriptive
b. Subscripted
c. Predictive
d. prescriptive

21. Which of the following library in python is used for plotting graphs and visualization?
a. Pandas
b. Jumpy
c. Matplotlib
d. None of Above

22. Which of the following command is used to install pandas?
a. pip install pandas
b. install pandas
c. pip pandas
d. None of Above

23. A_____ is a one dimensional Array?
a. DataFrame
b. Series
c. Both of Above
d. None of Above

24. The data label associated with a particular value of Seriesis called_____ ?
a. Data value
b. Index
c. Value
d. None of Above

25. Which of the following command s used to create a series ?
a. createSeries()
b. series()
c. create_series()
d. Series()

26. How many values will be there in array1, if given code is not returning any error?

series4 = pd.Series(array1, index = ["Jan", "Feb", "March" "April"])

a. 1
b. 3
c. 4
d. 2

27. Which of the following command s used to create a series ?

a. createSeries()
b. series()
c. create_series()
d. Series()

28. A _____ is a two dimensional labelled data structure ?

a. DataFrame
b. series
c. List
d. Tuple

29. In given code dataFrame df1 has ___rows and ____columns?

```
Import pandas as pd
Dict = {"Name":["Amit" , "Raj" , "Dev"], "RollNo":[1,2,3]}
Df1 = pd.dataFrame(dict)
```

a. 3,3
b. 2,3
c. 3,2
d. None of Above

30. Which of the following attribute of DataFrame is used to display all the values from DataFrame?

a. values

b. Values
c. val
d. Val

31. What is Numpy?
a. A data analysis library
b. A matrix manipulation library
c. A numerical computation library
d. A data visualization library

32. What is the function to create a Numpy array?
a. numpy.array()
b. numpy.asarray()
c. numpy.create_array()
d. numpy.make_array()

33. What is the method to find the data type of a Numpy array?
a. ndarray.item_type
b. ndarray.type
c. ndarray.data_type
d. ndarray.dtype

34. What is the function to create a Numpy array of zeros?
a. numpy.zeros_array()
b. numpy.zeros()
c. numpy.empty_array()
d. numpy.create_zeros()

35. What is the function to create a Numpy array of ones?
a. numpy.ones()
b. numpy.one_array()
c. numpy.create_ones()
d. numpy.full_array()

36. What is the method to reshape a Numpy array?

a. ndarray.reshape()
b. ndarray.resize()
c. ndarray.shape()
d. ndarray.modify_shape()

37. What is the method to find the sum of elements in a Numpy array?
a. array.sum()
b. numpy.sum()
c. array.add()
d. numpy.total()

38. Which of the method is used to check the dimension of an array ?
a. Shape
b. dim
c. shape
d. Dim

39. Which of the following is not an argument of arrange method?
a. start
b. stop
c. end
d. add

40. Which of the following is two dimensional array ?
a. [[1,2,3],[3,4,5]]
b. [1,2,3][3,4,5]
c. [1,2,3,4,56]
d. None of above

41. Which of the following method is use to create a scaler value array?
a. once_like

b. full
c. eye
d. diag

42. Which of the following method is use make the values static of random values array?
a. random. choice ()
b. random.randint()
c. random.seed()
d. random.random()

43. Which of the following is the method to change the data type of an array?
a. dtype
b. datatype
c. Astype
d. astype

44. Which of the following is the method multiply two arrays?
a. mult
b. multi
c. dot
d. Dot

45. Numpy package is cable to do the fast operations on array?
a. False
b. True
c. None of Above
d. Both a and b

46. Numpy array can be ?
a. Indexed
b. Sliced
c. Iterated
d. All of mentioned above

47. Observe the following code and identify what will be the outcome?

```
import numpy as np
a = np.array([1,2,3,4,5,6])
print(a)
```

a. [1,2,3,4,5]
b. [0,1,2,3,4]
c. [1,2,3,4,5,6]
d. [0,1,2,3,4,5]

48. What can be the output of following code?

```
import numpy as np
x = numpy.randint(100)
print(x)
```

a. All of below
b. 56
c. 98
d. 12

49. What are attributes of numpy array ?
a. shape, dtype ,ndim
b. objects ,type, list
c. Unicode, vectorization
d. All of above

50. What will be printed?

```
Import numpy as np
a = np.array([1,2,3,5,8])
b = np.array([0,3,4,2,1])
```

```
c = a+b
c = c*a
```

a. 7
b. 21
c. 12
d. 32

51. If a dimension is given as ___ in reshaping operation, the other dimensions are automatically calculated?
a. Zero
b. One
c. Negative One
d. Infinite

52. How to convert numpy array into list ?
a. array.list()
b. array_list()
c. list.array()
d. list(array)

53. Full function requires two arguments minimum to pass in it.?
a. False
b. True
c. None of Above
d. All of above

54. What is the correct syntax to print number 8 from the array below?
arr = np.array([1,2,3,4,5],[3,4,8,2,3])
a. print(arr[2,2])
b. print(arr[1,2])
c. print(arr[2,1])
d. None of Above

55. What is the correct syntax to print the numbers [3,4,5] from the array below?

```
arr = np.array([1,2,3,4,5,6,7])
```

a. print(arr[2:6])
b. print(arr[2:5])
c. print(arr[3:6])
d. print(arr[2:4])

56. Which of the correct syntax to print last 4 number from the array below?

```
arr = np.array([1,2,3,4,5,6,7])
```

a. print(arr[:4])
b. print(arr[4:])
c. print(arr[3:])
d. print(arr[4])

57. Which syntax you will use to print every other item in array?

```
arr = np.array([1,2,3,4,5,6,7])
```

a. print(arr[::2])
b. print(arr[0:,step=2])
c. print(arr[1:3:5:7])
d. None of Above

58. What is the correct method to to sort the element of an array?
a. orderby()
b. order()
c. sort()
d. Sort()

59. Which of the correct syntax to subtract the numbers from array to numbers from array2 ?

a. np.sub(arr1,arr2)
b. np.subtract(arr1,arr2)
c. np.min(arr1,arr2)
d. np.minus(arr1,arr2)

60. Which of the correct method to round decimal numbers in numpy ?

e. np.fix()
f. np.round()
g. np.Round()
h. np.trunc()

Course Name: DISCRETE MATHEMATICAL STRUCTURES

Course Outcome

CO1- Understand the notion of mathematical thinking, mathematical proofs, and algorithmic thinking, and be able to apply them in problem solving

CO2- Understand the basics of discrete probability and number theory, and be able to apply the methods from these subjects in problem solving.

CO3- Use effectively algebraic techniques to analyze basic discrete structures and algorithms.

Printed Pages: **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23
B. Tech. (Hons.) CS, Year II, Sem IV

BCSC0058: DISCRETE MATHEMATICAL STRUCTURES

Time: 2 Hours **Maximum Marks: 15**

Attempt All the Questions.

Section – A

Attempt All Questions 1 X 3 = 3 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	According to the survey made among 200 students, 140 students like cold drinks, 120 students like milkshakes and 80 like both. How many students like at least one of the drinks?	1	2	A	C
2.	f is a function defined by $f(x)=\begin{cases}2x+4 & x\leq 2\\ 2x-1 & x>2\end{cases}$ Find f(0), f(2) and f(4)	1	1	R	F
3	Two finite sets have m and n elements. The total number of subsets of the first set is 56 more the total number of subsets of the second set. Find the values of m and n .	1	3	An	C

Section – B

Attempt All Questions 2 X 3 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
4	A single pair of rabbits (male and female) is born at the beginning of a year. Assume that 1) rabbit pairs are not fertile during their first month of life but every month after that they give birth to two new male/female pairs at the end of each month and 2) no	2	3	An	M

	rabbits die. Find a recurrence relation for the number of male-female rabbit pairs at the end of month n. How many rabbits will there be at the end of 12 months?				
5	Let X= {1,2,3,4,5,} and relation R= {(x, y)/x>y}.Draw the graph of 'R' and also give its matrix.	2	1	R	F
6	In how many ways can 8 different dolls be packed in 5 identical gift boxes such that no box is empty if any of the boxes hold all of the toys?	2	2	U	M

Section – C

Attempt All Questions 3 X 2 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
7	Prove that for any positive integer number n, $n^3 + 2n$ is divisible by 3	3	2	A	P
8	Solve the following recurrence relations using generating functions: $a_r = 3a_{r-1} + 2, \quad r \geq 1, \quad a_0 = 1.$	3	1	U	P

Course Name: **English for Professional Purposes II**

Course Outcomes

CO1- Students will be able to communicate more accurately, briefly and clearly.

CO2- Students will be able to create better PowerPoint presentations.

CO3- Students will develop confidence in communication at workplace.

CO4- Students will be able to understand nuances of English Grammar and rectify them.

CO5- Students will be more conscious in academic and professional communication, written and spoken.

Printed Pages: 10 **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23

B.Tech (all branches), Biotech, IV Semester

Subject Code & Subject Name- _BELH0004 English for Professional Purposes II

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15

No.	Detail of Question	Marks	CO	BL	KL
1	**Rearrange the following sentences to form a coherent paragraph. The first and the last sentences are I and VI, and the four in between are labelled A, B, C and D. Choose the most logical order of these four sentences from among the four given choices to construct a coherent paragraph from sentences I to VI** **1.1.** I. Making people laugh is tricky. A. At times, the intended humour may simply not come off. B. Making people laugh while trying to sell them something is a tougher challenge, since the commercial can fall flat on two grounds.	3	1	UA	C

CV

C. There are many advertisements which do amuse but do not even begin to set the cash tills ringing.
D. Again, it is rarely sufficient for an advertiser simply to amuse the target audience in order to reap the sales benefit.
VI. There are indications that in substituting the hard sell for a more entertaining approach, some agencies have rather thrown out the baby with the bath water.
a) DACB b) ADCB c) ABCD d) BADC e) CBAD

1.2.
I. Security inks exploit the same principle that causes the vivid and constantly changing colours of a film of oil on water.
A. When two rays of light meet each other after being reflected from these different surfaces, they have each travelled slightly different distances.
B. The key is that the light is bouncing off two surfaces, that of the oil and that of the water layer below it.
C. The distance the two rays travel determines which wavelengths, and hence colours, interfere constructively and look bright.
D. Because light is an electromagnetic wave, the peaks and troughs of each ray then interfere either constructively, to appear bright, or destructively, to appear dim.
VI. Since the distance the rays travel changes with the angle as you look at the surface, different colours look bright from different viewing angles.
a)ABCD b) BADC c) BDAC d) DCAB e) BACD

	1.3. I. According to recent research, the critical period for developing language skills is between the ages of three and five and a half years. A. The read-to child already has a large vocabulary and a sense of grammar and sentence structure. B. Children who are read to in these years have a far better chance of reading well in school, indeed, of doing well in all their subjects. C. And the reason is actually quite simple. D. This correlation is the highest yet found between home influences and school success. VI. Her comprehension of language is therefore very high. a) DACB b) ADCB c) ABCD d) BDCA e) CBAD				
2	**Fill in the blanks:** 2.1 The canteen provides the workers _____ meals. A. with B. at C. for D. to 2.2 Many generations of people have fought ..... freedom. A. with B. for C. about D. at 2.3 If you only think .....failure, you'll never take any risks. A. with B. for C. about D. of	3	4	A	C
3	**Spot the error, if any:** 3.1. The feeling of anticipation and excitement (A)/ was in the air (B)/ as the house manager took the stage (C)/ and says, "Ladies and gentleman, thank you for your	3	1	A	C

	support!". (D)/ NO ERROR (E) 3.2. To tell you the truth (A)/I would be glad to retire from business altogether (B)/ considering that I was now past seventy years (C)/ of age and have a daughter to succeed me. (D)/NO ERROR (E) 3.3. I take this course (A)/ to improve my grammar (B)/ as this will provides (C)/ basic knowledge of the language. (D)/ NO ERROR (E)				
4	**Fill in the blanks:** 4.1. Many students rely …….. Internet to conduct research. A. on B. for C. about D. of **Choose the correct alternative for the underlined part of each sentence.** 4.2. A bit of fragrance **is always clinging to** the hand that gives roses. A. will always cling B. always clings C. was always clinging D. has always been clinging E. no correction required 4.3. Pandit Jawaharlal Nehru **make** his famous midnight speech, India's "Tryst with Destiny," on the night of 14-15 August 1947. A. had made B. made C. has made D. have made E. was making	3	1	An	M

5	**Fill in the blanks:** 5.1. She has written a _____ of her encounters with W.H. Auden over the years. A. agenda B. memo C. memoir D. email 5.2. His defeat in the world championship led to a long period of gloomy. _____. A. introspection B. paradox C. combustion D. simulation 5.3. A style in which a writer makes a display of his knowledge is known as.. A. Pedantic B. Verbose C. Telegraphic D. Ornate	3	3	U	C

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	6.1. The underlined words in the following sentences are emphasized. What variation in the meaning will the emphasized words bring to these sentences? A. **I** asked you to read it aloud. B. I asked **you** to read it aloud. C. I asked you to read it **aloud**. 6.2. Differentiate between content words and function words. ***OR*** What do you mean by 'sense groups'? How do	3+2=5	3	An	M

	'sense groups' make it easier for the speaker to convey the message accurately and for the listener to comprehend the message correctly?				
7	**7.1. Improve the following sentences so as to make them suitable for professional writing:** A. He dropped out of school on account of the fact that it was necessary for him to help support his family. B. The subjects that are considered most important by students are those that have been shown to be useful to them after graduation. C. I have got an email from Mrs. Reena Chauhan. D. A.student's beliefs about feminism may be based on what he has heard in the popular media. **7.2. Discuss the possible meanings of the following sentences:** A. Dr Mishra's examination was a long one. B. The case was about his wife's fatal shooting. C. He is a foreign language teacher.	2+3=5	5	C	C
8	Write an essay on any one of the following topics (300 words): The Worst Day of my Life India in 21st Century ***OR***	5	5	UC	M

Read the essay given below and answer the following questions:

A. Do you think this essay is well organized? Illustrate.

B. Identify the transition markers and see whether they have been used correctly or not.

Agriculture in India

Agriculture is the oldest industry in India. Old books and scriptures also mention it. There are gods and goddesses and prayers to them for boons to agriculture. In India, it has also been seen as the best occupation. Trade has been seen as the second best, and being servant of others has been seen as the last resort, better only than begging.

For centuries, it has been the backbone of the Indian economy. Numbers have fallen now, but until recently it used to employ over 80 percent of India's total population. Agriculture here was so good that barring some exceptions, India produced more than enough for its own people. Along with minerals and metals, artefacts and textiles, food grains have been India's principal export ever since there has been export of anything from India. Actually, agriculture here was so vibrant that the only people the British took out of India for their help, other than of course soldiers to fight for them, are the farm labourers who brought them prosperity through

farming of sugarcane and other commercial crops.

India grows a variety of crops, larger than perhaps any other country does. I cannot imagine any agricultural product that is produced elsewhere but not in India. It produces a variety of cereals from rice to wheat, maize, millet, barley, corn, etc. Then there are over 15 different kinds of protein rich pulses. Of fruits and vegetables there is such a large variety, that even cataloguing them all here in this short essay is not possible. Commercial agriculture has also been very popular in India. Tea, coffee and tobacco were brought to India by the Europeans, but it has always had sugarcane and spices which have brought Europeans and other foreigners to India, and which even today brings valuable foreign exchange. Now Indian tea and coffee are big foreign exchange earners. The best thing about Indian industry is that until recently it used only eco-friendly methods of farming, using animals and their dung and vegetation to sustain the productive capacity of soil. Animals have been both used and sustained by Indian agriculture.

Everyone values the importance of agriculture in the economy of India. Beginning with the British, many universities and research institutions have been established in India to improve the productivity and conditions of farming and farmers have been created in India. Union and all state governments have ministries

and departments of agriculture. Its place in the economy is unquestionable.

Yet, in recent years, the agriculture sector has not attracted the kind of manpower it should have. Best of young people hardly any longer come to agriculture for their occupation. Besides, the government has done little to bring modern methods of farming affordable and sustainable in India. Old methods of irrigation and flood control have disappeared. As a result, Indian agriculture has been plagued with problems like increasing suicides by farmers, debt-burdened farmers, successive years of droughts and floods, and India has done nearly nothing to change this situation.

We must make tillers, sprayers, sprinklers, power generators, harvester, thrasher, etc for our own needs such that they do not take a lot of power neither do they cost a lot of money and work well on both small and big plots of land. Agriculture marketing must be more business-like. Famers do not need subsidies, but they should be allowed to charge competitive prices. If the price of soap is not regulated, why should the price of sugarcane be regulated! If we can do these things, then as the famous agriculture scientist, Dr M S Swaminathan, says, Indians can feed the world.

With these measures, Indian agriculture is still capable of regaining its old glory and power. It will attract talented and enterprising people and

(9/

	will reward them with returns worthy of respect. Earth is seen as mother in India. And mother can never fail. All that is required is giving agriculture its due place. Rewards can be incalculable.				

Course Name: Database Management System

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real life application, with constraints and keys using SQL.
CO5: Select the information from a database by formulating complex queries in SQL.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

Printed Pages: 03 **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23

B. Tech. (CSE/AIML/CCV/DA/IIoT/CSF/CSED), Year: II, Semester: IV

Subject Code & Subject Name- BCSC1003 Database Management System

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
1. Attempt both sections.
2. Use only blue pen to answer.
3. No B sheet will be given.

Section – A

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider a scenario in which you are appointed as a database administrator in an organization. What roles and responsibilities you are supposed to perform? Explain it.	3	CO1	A	P
2	Define Entity integrity, Domain integrity and Referential integrity in details.	3	CO3	U	F
3	What do you mean by Weak entity type? Name the relationship that exist between a strong entity and a weak entity.	3	CO3	U	F
4	Consider the Relation R(WXYZ) and Functional Dependencies given below: FD: $\{W \rightarrow X, X \rightarrow Y, Y \rightarrow W\}$ Identify Candidate Key(s) in this relation.	3	CO4	A	P
5	Explain the following terms: (a) Participation constraints in ER model (b) Generalization & Specialization in EER model (c) Super Key	3	CO3	U	F

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider the following relation: ***STUDENT*** *(R_NO, S_NAME, S_COURSE, CPI, ADDRESS, EMAIL_ID)* Take datatype and range as per the suitability of this question. Write the SQL syntax for following queries: (a). Write the syntax to create STUDENT table. (b). Insert a tuple in STUDENT table. (c). Delete the record of Student who live in 'Agra'. (d). List the details of the students in order of the descending of CPI. (e). List the student names those are starting with 'A' and with five characters.	5	CO4	A	P
7	Consider a case in which you are given the following requirements for a simple database for the National Hockey League (NHL): • The NHL has many teams, • Each team has a name, a city, a coach, a captain, and a set of players, • Each player belongs to only one team, • Each player has a name, a position (such as left wing or goalie), a skill level, and a set of injury records, • A team captain is also a player, • A game is played between two teams (referred to as host_team and guest_team) and has a date (such as March 03rd, 2023) and a score. Construct a clean and concise ER diagram for the NHL database. List your assumptions and clearly indicate the cardinality mappings in your ER diagram.	5	CO3	A	P

8	Consider the following schema: DEVELOPER (**DEV_ID**, DEV_NM, ADDRESS) MANAGER (**MG_ID**, MG_NM, MG_PROJ_NM) COMPANY (**DEV_ID, MG_ID**, COMP_NAME) Attributes written in bold letters are prime attribute(s). Write Relational Algebra expression for the following queries: (a) Find the name of developers who live in 'Chennai'. (b) Find the name of Manager who is working on Project 'ABC'. (c) Find the name of manager who is managing Developer 'Arvind' for project 'XYZ'. (d) Insert a new tuple into relation MANAGER. (e) Delete records of Developers whose address is 'Pune'.	5	CO3	A	P

Course Name: Operating Systems

Course Outcome

CO1- Understand the classification of operating system environment.
CO2- Understand the basic of process management.
CO3- Apply the concept of CPU process scheduling for the given scenarios.
CO4- Illustrate the process synchronization and concurrency process in operating system.
CO5- Analyze the occurrence of deadlock in operating system.
CO6- Describe and analyze the memory management and its allocation policies.
CO7- Understand the concepts of disk scheduling

Printed Pages:2 **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23
B. Tech. (CSE), II Year, IV Semester
Operating Systems (BCSC0004)

Time: 2 Hours **Maximum Marks: 30**

Note: Attempt all questions.

Section – A 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider the following set of processes, assumed to have arrive at time 0. Consider the CPU scheduling Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order P1, P2, P3, P4 Processes | P1 | P2 | P3 | P4 Burst time (in ms) | 8 | 7 | 2 | 4 If the time quantum for RR is 4 ms, then the find the absolute value of the difference between the average turnaround time (in ms) of SJF and RR (round off to 2 decimal places).	3	3	A	P
2	Discuss the following operating system types with their pros and cons: (a) Real time systems (b) Multiprocessor systems	3	1	U	F
3	What is a process? How a process is represented in a memory, explain with a diagram?	3	2	U	C
4	What is Inter-process communication (IPC)? Explain the two models used for the IPC.	3	4	R	F

5	What is a critical section problem? What are the different requirements that must satisfy a solution to the critical section problem. Also, in below algorithm, check whether these requirements are satisfied or not, if processes execute their respective codes where M is a Boolean array of two elements.
	
	Process P_0
	M[0]=true;
	while(M[1]==true);
	Critical section;
	M[0]=false;
	Remainder section
	
	Process P_1
	M[1]=true;
	while(M[0]==true);
	Critical section;
	M[1]=false;
	Remainder section
		3	4	E	M

Section – B 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is context switching? Explain steps involved in it with the help of a diagram. Also, explain the queue diagram representation of process scheduling.	5	2	U	C
2	What is aging? Consider a set of 5 processes whose arrival time, CPU time needed and the priority are given below: Process / Arrival Time(in ms) / CPU time needed / Priority P1 / 0 / 10 / 5 P2 / 0 / 5 / 2 P3 / 2 / 3 / 1 P4 / 5 / 20 / 4 P5 / 10 / 2 / 3 calculate the average waiting and turnaround time using pre-emptive priority scheduling. Assume 1 is the highest priority.	5	3	A	P
3	Discuss the usage of Semaphore? Distinguish between Binary and Counting Semaphore. Also, provide the solution for the Dining Philosopher Problem using the same.	5	4	U	P

Course Name: Cloud and Business Process Management

Course Outcome

CO1-Understand basics of cloud Storage systems.
CO2-Explian the technologies and approaches for the business-related issues.
CO3-Understand the operation view and services catalog of cloud management.
CO4-Understand the concepts of VPM cloud computing.
CO5-Design process interaction interfaces for business users.

Printed Pages:01 **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23
B.Tech (CCV), Year-II, Semester-IV
BCSE 0508 Cloud and Business Process Management

Time: 2 Hours **Maximum Marks: 30**

Instruction for students: Attempt all questions of a section at one place.

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Who are the major participants in the cloud computing platform as a service layer? Use an appropriate example to describe the on-demand self-service feature.	3	2	An	C
2	Why is BPM meaningful? Give specific examples of business agility.	3	3	U	P
3	Describe BPM. Explain with the help of its life cycle.	3	3	R	F
4	Which three services are offered by the ANEKA Cloud Architecture's middleware container?	3	1	R	F
5	What does the cloud architecture's "Unified Layer" mean? Describe using a diagram of the cloud architecture and the functionality of virtualization.	3	4	R	F

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Given all of the various aspects of cloud computing, why is cloud management so important? Explain with the help of the diagram.	5	1	U	C
7	What is cloud provisioning, and how does it function? Examine several provisioning model types. Write short notes on orchestration and automate.	5	1	R	M
8	The BPM evaluation phase makes use of available information to access. clarify this statement in depth. Include the administrator and shareholder in your response as well. What other kinds of business processes are there?	5	4	A	P

Course Name: Applied Statistical Analysis

Course Outcome

CO1-Understand the different variables measurement and scalling.
CO2-Analyze the different central tendency operations.
CO3-Demonstrate the different ways to represent the data.
CO4-Analyze the hypothesis testing using different estimation and distribution methods.
CO5-Analyze the correlation and regression for the real world problem data.
CO6- Analyze the data using the advanced techniques like , Chi-squared goodness-of-fit test, Chi- square test of independence, the sign test and Mann-Whitney test.

Printed Pages: 3 **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23
B.Tech (CSE (DA), II[nd] Year, IV Semester
Subject Code & Subject Name- BCSE0552 & Applied Statistical Analysis

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	In a factory, there are 100 skilled workers, 250 semi-skilled and 150 unskilled workers. It has been observed that on an average a unit length of a particular fabric is woven by a skilled worker in 2 hours, by a semi-skilled worker in 3 hours and by an unskilled worker in 4 hours. After a training Of two years, the semi-skilled workers are expected to become skilled and unskilled workers to become semi-skilled. How much less time will be required after two years of training for weaving the unit length of fabric by an average worker?	3	2	A	C
2	Cities A, B and C are equidistant from each other. A motorist travels from A to B at 0 km/h; from B to C at 40 km/h and from C to A at 50 km/h. Determine the average speed for the entire trip.	3	2	A	C
3	For a frequency distribution of marks in Econometrics of 200 candidates (grouped in intervals of 0– 5, 5– 10, ... etc.), the mean and the standard deviation were found to be 40 and 15,respectively. Subsequently, it was found that the score 43 was wrongly taken as 53 in obtaining the frequency distribution. What should be the corrected mean and the standard deviation?	3	3	E	C
4	In a frequency distribution, coefficient of skewness based on quartiles is 0.6. If the sum of the upper and	3	3	A	C

	lower quartiles is 100 and the median is 38, find the value of the lower and upper quartiles.				
5	A, B and C bidding for a contract. It is believed that A has exactly half a chance that B has; B, in turn, has 4/5th as likely as C has to gain the contract. What is the probability for each to win the contract?	3	2	A	C

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No	Detail of Question	Marks	CO	BL	KL
6	A website displays 10 advertisements and the revenue generated by the website depends on the number of visitors to the site clicking on any of the advertisements displayed on the website. The data collected by the company has revealed that out of 2500 visitors, 30 visitors clicked on 1 advertisement, 15 clicked on 2 advertisements, and 5 clicked on 3 advertisements. Remaining did not click on any of the advertisements. Calculate (a) The probability that a visitor to the website will click on an advertisement. (b) The probability that the visitor will click on at least two advertisements. (c) The probability that a visitor will not click on any advertisements.	5	3	E	C
7	Time between failures (in hours) of a wire cutter used in a cookie manufacturing oven is given in Table1 The function of the wire-cut is to cut the dough into cookies of desired size. 2 22 32 39 46 56 76 79 88 93 3 24 33 44 46 66 77 79 89 99 5 24 34 45 47 67 77 86 89 99 9 26 37 45 55 67 78 86 89 99 21 31 39 46 56 75 78 87 90 102 (a) Calculate the mean, median, and mode of time between failures of wire-cuts. (b) The company would like to know by what time 10% (ten percentile or $P10$) and 90% (ninety	5	2	E	P

	percentile or $P90$) of the wire-cuts will fail? (c) Calculate the values of $P25$ and $P75$.				
8	Calculate the Pearson's measure of skewness on the basis of mean, mode and standard deviation: Mid-value 14.5 15.5 16.5 17.5 18.5 19.5 20.5 21.5 Frequency 35 40 48 100 125 87 43 22	5	3	A	F

Course Name:

Course Outcome

CO1- Understand the fundamental of Network Security
CO2- Demonstrate the principle of network security control protocol and device.
CO3- Understand the basic concepts of physical and host security, and secure IDS configuration & Management
CO4- Explain the network security by using VPN and wireless technologies.
CO5- Explain the concepts of network traffic monitoring and analysis.
CO6: Conceptualize the risk management framework and vulnerability management.
CO7: Explain the concepts of data backup, recovery tools and incident response management.

Printed Pages: **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23
B.Tech (CSF), II, IV
Subject Code & Subject Name- BCSE0602 IT Network Security

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain the type of firewall in details. Or Explain the physical security control?	3	3	R	C
2	Why VPN is used in network. Explain the advantage of VPN over the normal Network. Or Explain the host security baseline?	3	4	U	C
3	Explain the CIA model of security? Or Explain the auditing windows registry	3	1	R	C
4	How we can increase the security of data with the help of IP Sec explain in detail. Or Explain the Network Address Translator and what is its requirement in the network?	3	5	A	F
5	Explain the threats and vulnerabilities in the network? Or Why the data backup policy and method?	3	6	R	P

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL

6	Perform the man in middle attack on the Diffie Hellman Key exchange protocol. Let two public prime no g=3, n=5, secrete value of sender x=2 and secrete value of receiver is y=4	5	6	An	P
7	Explain the Digital Signature with the RSA algorithm. If P=7, Q=17 E=5 and plain text is 10 in RSA algorithm take these values and explain the Digital Signature with the RSA algorithm. Find out the vale of D and cipher text.	5	5	An	P
8	Intrusion Detection System is need of network security in present time explain with an example	5	2	U	P

Course Name:

Course Outcome

CO1- Understand the way in which Intelligent System is methodologically distinctive while at the same time is an interdisciplinary field where established fields of research- including Psychology, Computer Science, Linguistics, Neuroscience, and Philosophy-contribute to the overall approach.

CO2- Develop skills in analyzing, interpreting, and assessing the empirical data and research techniques that contribute to intelligent system

CO3- Understand central modeling techniques in intelligent, including traditional computational approaches, neural network/deep learning approaches, and dynamical approaches.

CO4- Learn to communicate technical work in particular research areas in writing and in oral presentation to an interdisciplinary audience.

CO5- Able to use critical thinking to evaluate and interpret evidence, and to apply psychological concepts, theories, and research findings to individuals, social, and cultural issues.

Printed Pages: **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23
B.Tech (AIML), 2nd Year, IVth Sem.
Subject Code & Subject Name- BCSE0702Algorithm for Intelligent Systems and Robotics

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	State three stratum of Human Intelligence	3	CO1	U	C
2	What is fluid intelligence?	3	CO2	A	P
3	What are the syntax and semantic of propositional logic?	3	CO3	A	P
4	What is clause in propositional logic?	3	CO4	U	D
5	What is particle swarm optimization (PSO) ?	3	CO5	E	DI

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	What is resolution. Give an example of resolvent?	5	CO1	U	C
7	Explain Prenex normal form in first order logic	5	CO3	E	C
8	Give an example of skolen standard form	5	CO3	U	P

Course Outcome

CO1- Learn good principles of algorithm design.

CO2- Apply the algorithms and design techniques to solve problems.

CO3- Analyze the complexities of various problems in different domains and estimate their worst-case, average-case and best case behavior.

CO4- Discuss various searching, sorting, and graph traversal algorithms.

CO5- Know how to design algorithms using the divide-and-conquer. Dynamic programming, greedy approach strategy and recite algorithms that employ these strategies.

Printed Pages: **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23
B. Tech. (Hons.) CS, 2nd Year, Semester (IV)
Subject Code & Subject Name- BSSC 0059 (Algorithms Design & Analysis)

Time: 2 Hours **Maximum Marks: 15**

Instruction for students:

1. Attempt all questions
2. Do not switch between section A and Section B, first, solve one section, then move to second
3. Use only a blue pen to answer
4. No B sheet will be given

Section – A

Attempt All Questions 1 X 3 = 3 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Write the recurrence relation to find the GCD of given two numbers.	1	2	U	C
2	Write the recurrence relation of outplace sorting algorithm and also analyze its time complexity.	1	3	An	C
3	Write the properties of Binomial Trees.	1	5	R	F

Section – B

Attempt All Questions 2 X 3 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
4	What do you understand by asymptotic analysis of an algorithm? Explain different types of asymptotic notations.	2	1	R	F
5	Solve a recurrence relation using iterative method. $T(n) = 16\,T(n/4) + n^3$	2	2	E	C
6	Write and explain an unstable sorting algorithm using DAC strategy. Also analyze the time complexity.	2	4	An	C

Section – C

*Attempt **All** Questions* 3 X 2 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
7	Perform an insertion operation using a suitable algorithm in B Tree on the following data set: 7, 4, 13, 8, 12, 3, 6, 9, and 11. Degree is given 2.	3	5	A	P
8	What are the properties of red-black tree? Design a suitable tree with satisfies all the properties of red-black tree. A= "ALGORITHMS"	3	5	A	P

Course Name: Machine Learning Using Python

Course Outcome

CO1-Apply the basic concepts of machine learning.
CO2- Analyze the concepts of regression and classification methods.
CO3- Apply the concepts of supervised machine learning algorithms: KNN, Decision Tree, and SVM
CO4- Design unsupervised learning-based solutions.
CO5- Analyze the concepts of artificial neural network methods.
CO6-Design the applications of machine learning algorithms.

Printed Pages: 02 **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23
B. Tech. (Hons.) CS, II Year, IV Semester
BCSE0450, Machine Learning Using Python

Time: 2 Hours **Maximum Marks: 15**

Section – A

Attempt All Questions 1 X 3 = 3 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Can all problems be solved using machine learning? Write your response in brief.	1	1	A	C
2	Write down the hypothesis, parameters, and cost function of linear regression.	1	2	U	F
3	Can the performance of a learning model be improved? If yes, write down the reason.	1	2	R	F

Section – B

Attempt All Questions 2 X 3 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
4	While predicting win-loss of teams in World Cup Football using a classification model, following are the data recorded: a. Correct predictions – 73 wins, 13 losses b. Incorrect prediction – 9 wins, 5 losses. Calculate the accuracy, precision, recall and F-measure of the model	2	2	A	P
5	Find the slope of the graph where the lower point on the line is represented as (-3, -2) and the higher point on the line is represented as (2, 2).	2	2	A	P
6	What happens when there is no clear Hyperplane in SVM? How is it handled?	2	3	AN	C

Section – C

Attempt All Questions 3 X 2 = 6 Marks

<table>
<tr><th>No.</th><th>Detail of Question</th><th>Marks</th><th>CO</th><th>BL</th><th>KL</th></tr>
<tr><td>7</td><td>Compute parameters of a linear regression model.

<table><tr><td>Number of hours (X)</td><td>10</td><td>9</td><td>2</td><td>15</td><td>10</td><td>16</td><td>11</td><td>16</td></tr><tr><td>Risk Score on a scale of 0-100 (Y)</td><td>95</td><td>80</td><td>10</td><td>50</td><td>45</td><td>98</td><td>38</td><td>93</td></tr></table>a. Find the regression line Y = A.X + B.
b. Use the regression line as a model to estimate the Risk Score on 12 hours of driving</td><td>3</td><td>3</td><td>A</td><td>P</td></tr>
<tr><td>8</td><td>We have data from the questionnaires survey (to ask people opinion) and objective testing with two attributes (acid durability and strength) to classify whether a special paper tissue is good or not. Here are six training samples.
<table><tr><th>X1 = Acid Durability (seconds)</th><th>X2 = Strength (kg/square meter)</th><th>Y = Classification</th></tr><tr><td>1</td><td>4</td><td>Good</td></tr><tr><td>7</td><td>7</td><td>Bad</td></tr><tr><td>7</td><td>4</td><td>Bad</td></tr><tr><td>3</td><td>4</td><td>Good</td></tr><tr><td>2</td><td>5</td><td>Good</td></tr><tr><td>1</td><td>3</td><td>Bad</td></tr></table>Now the factory produces a new paper tissue that pass laboratory test with X1 = 3 and X2 = 7. Without another expensive survey, can we guess what the classification of this new tissue with the help of KNN algorithm. Solve it for both K=3 and K=5.</td><td>3</td><td>3</td><td>A</td><td>P</td></tr>
</table>

Course Name: B.Tech Hon.

Course Outcome

CO1-The course aims to develop an understanding to creativity and innovation.

CO2-It harness creativity and change leadership to develop professional and organizational skills.

CO3-It develops contemporary and radical ways of thinking to develop problem solving skills with new possibilities.

CO4-It structures innovation leadership skills to inspire and motivate others by some proven frameworks to apply innovative ideas to design new practices.

CO5- It develops innovation management skills.

Printed Pages: **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23
B. Tech. (Hons.) CS, II Year, IV Semester
BELG-0001: Creativity and Innovation

Time: 2 Hours **Maximum Marks: 15**

Instruction for students:

Section A: word limit is 100 words

Section B: Word limit is 200 words

Section C: Draw design, diagrams or sketch; Word Limit is 300 words

Section – A

*Attempt **All** Questions* 1 X 3 = 3 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is the role of IFF in creativity and Innovation?	1	1	U	C
2	What is CIC mindset to bring new designs to solve the problems? Give two examples.	1	3	A	M
3	What is the importance of Idea Journal? Explain through real life examples.	1	4	E	F

Section – B

*Attempt **All** Questions* 2 X 3 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
4	Wild mind can bring innovation in creativity: explain with some examples which you experience in IoT based systems.	2	2	An	C
5	Explain the creativity by analyzing the revised Bloom's taxonomy pyramid?	2	1	U	M
6	Discuss the four categories of creativity with examples.	2	1	R	F

Section – C

*Attempt **All** Questions* 3 X 2 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
7	Create a class room best suited for learning and development using the digital transformations.	3	5	C	M
8	What are the creative and innovative changes you would like to make in what's app for better consumer service?	3	4	A	P

Course Name: Software Engineering

Course Outcome

- CO1: Understand the basic concepts of software engineering.
- CO2: Apply software processes to solve real world problems.
- CO3: Estimate the cost, effort and schedule of software using COCOMO Model.
- CO4: Analyze the software design techniques (structure chart, SDM, sequence diagram).
- CO5: Understand the basic concepts of OO analysis and design.
- CO6: Develop the test cases to validate the software.
- CO7: Understand the basic models of software Quality and maintenance.

Printed Pages: 2 **University Roll No.....**

Mid Term Examination, Even Semester 2022-23
B. Tech (CCV, DA, IIOT, AIML, CSF, CSE)
II Year, IV Semester

Subject Code & Subject Name- BCSC0009 & Software Engineering

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is the aim of Software Engineering? What are the components of Software? Discuss in Detail.	3	1	U	F
2	Explain how the principles underlying agile methods lead to the accelerated development and development of software. Why Software Industry Moving from Traditional models to Agile Software development model.	3	1	R	F
3	Assume that you are the technical manager of a software development organization. A client approached you for a software solution. The problems stated by the client have uncertainties which lead to loss if it not planned and solved. Which software development model you will suggest for this project-justify. Explain the model with its pros and cons and neat sketch.	3	2	U	A
4	Discuss the elicitation technique which promotes the creative thinking and new ideas.	3	2	R	F
5	Consider the following activities of Restaurant System: Customer Order, Serve Product, Collect Payment,	3	5	An	C

	Produce Product, Store Product, Order Raw Materials, Pay for Raw Materials, and Pay for Labor. Consider the above scenario and create the Use Case Diagram.				

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider a project of GLA University Management System with following information No. of I/P = 30 (low) No. of O/P = 42(high) No. of user Inquiries = 8(average) No. of files = 7(low) No. of external interfaces = 6(high) In addition to above, system requires moderate master file updating and significant code reusability and other CAF are treated as average. Calculate Effort, Development Time and Productivity and staff size of the project. Suppose the project is to be developed in C language. All cost drivers are nominal.	5	3	A	P
7	How modularity plays an important role in designing software? Differentiate coupling and cohesion? Explain different types of Module Coupling with the help of an example.	5	5	A	D
8	Create Level-0, Level-1 for online shopping website. The admin of the website can login as an admin and can manage the items of the company. Admin can manage the orders placed by customers, generates the monthly sales report. Users can login, search, place order, track order and online payment of purchased items.	5	5	An	C

Course Name: OPERATING SYSTEM AND CONCEPTS

Course Outcome

CO1: Understand the classification of operating system environment.
CO2: Understand the basic of process management.
CO3: Apply the concept of CPU process scheduling for the given scenarios.
CO4: Illustrate the process synchronization and concurrency process in operating system.
CO5: Analyze the occurrence of deadlock in operating system.
CO6: Describe and analyze the memory management and its allocation policies.
CO7: Understand the concepts of disk scheduling.

Printed Pages:02 **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23
B. Tech. (Hons.) CS, 2nd Year, VI Semester
BCSE0055 : Operating System and Concepts

Time: 2 Hours **Maximum Marks: 15**

Section – A

Attempt All Questions 1 X 3 = 3 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What inconveniences that a user can face while interacting with a computer system without an operating system? OR "Operating system is a resource manager". Justify this statement with suitable functionality of OS.	1	1	A	C
2	List the data fields associated with process control block.	1	2	R	F
3	Under what circumstances user level threads are better than the kernel level threads?	1	2	U	C

Section – B

Attempt All Questions 2 X 3 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
4	Explain time slicing? How its duration affects the overall working of the system?	2	1	A	P
5	What are the advantages and disadvantages of using the same system call for both files and devices? OR Discuss the transitional changes in process states diagram when blocked-suspended and ready-suspended states are included.	2	1	U	C

6	Explain microkernel and multithreading operating system design with advantages and disadvantages.	2	1	R	F

Section – C

*Attempt **All** Questions*

3 X 2 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
7	Distinguish between counting and binary semaphores. Show when does the semaphore definition requires busy waiting. Suggest a solution to overcome this problem. OR What are the advantages of inter-process communication? How communication takes place in a shared-memory environment? Explain.	3	4	U	P
8	Assume the following workload in a system: Process / Arrival Time / Burst Time P1 / 5 / 5 P2 / 4 / 6 P3 / 3 / 7 P4 / 1 / 9 P5 / 2 / 2 P6 / 6 / 3 Draw a Gantt chart illustrating the execution of these jobs using Round robin scheduling (with quantum =2) algorithm and also Calculate the average waiting time and average turnaround time.	3	3	An	P

Course Name: **Software Engineering Methodologies BCSC0057**

Course Outcome

CO1: Understand the basic concepts of software engineering.
CO2: Apply software processes to solve real world problems.
CO3: Estimate the cost, effort and schedule of software using COCOMO Model.
CO4: Analyze the software design techniques (structure chart, SDM, sequence diagram).
CO5: Understand the basic concepts of OO analysis and design.
CO6: Develop the test cases to validate the software.
CO7: Understand the basic models of software Quality and maintenance.

Printed Pages: 3 **University Roll No. .....**

Mid Term Examination, Even Semester 2022-23
B. Tech. (Hons.) CS, Year: II, Semester: IV
Subject Name & Code- Software Engineering Methodologies BCSC0057

Time: 2 Hours **Maximum Marks: 15**

Section – A

Attempt All Questions 1 X 3 = 3 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Why do we feel that characteristics of requirements play a very significant role in the selection of a life cycle model? Also Enlist various factors on which the management of software development is heavily dependent.	1	1	R	F
2	What is the objective of requirement validation? Also draw the requirement review process.	1	1	R	F
3	Why KLOC is not universally accepted as a measure for software effort estimation? Also Discuss the various characteristics of software.	1	3	U	C

Section – B

Attempt All Questions 2 X 3 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
4	Explain the software life cycle model in which the radial dimension of the model represents the cumulative costs. What are the advantages of this model?	2	2	U	C
5	What are crucial process steps of requirement engineering? Discuss the requirement elicitation technique in which the voice of customer is translated into technical requirements.	2	1	U	F
6	Define Structure Chart. Draw the structure chart for the following program by showing data and control flow.	2	4	E	M

```
main ( )
{
int: x, y;
x=0; y=0;
add(a[ ],n); mul(b[ ],n); div();
--------
--------
}
add(a[ ],n)
{ x=x+y;   y=y+10; }
mul(b[ ],n)
{ x=x+10; y=y+x; display(x,y); }
Display ()
{ Print("results"); }
```

Section – C

Attempt All Questions 3 X 2 = 6 Marks

No.	Detail of Question	Marks	CO	BL	KL
7	Consider a project with the following function units: 30 low external inputs, 40 high external outputs, 20 low internal logical files, 25 high external interface files, 15 average external inquiries. In addition to above, system requires moderate master file updating and significant code reusability and other CAF are treated as average. Suppose the project is to be developed in C language having size 128 SLOC/FP. All cost driver is nominal i.e 1.0. a. Calculate Effort, Development Time and Productivity of the project. b. Phase wise effort and development.	3	3	A	P
8	Consider the following activities of Restaurant System: Customer Order, Serve Product, Collect Payment, Produce Product, Store Product, Order Raw Materials, pay for Raw Materials, Pay for Labor. Consider the above scenario and create the following: **a.** Use Case Diagram **b.** DFD level-0, level-1.	3	2	An	P

Lifecycle Phase Values of μ_p

Mode & Code Size	Plan & Requirements	System Design	Detailed Design	Module Code & Test	Integration & Test
Organic Small S=2	0.06	0.16	0.26	0.42	0.16
Organic medium S=32	0.06	0.16	0.24	0.38	0.22
Semidetached medium S=32	0.07	0.17	0.25	0.33	0.25
Semidetached large S=128	0.07	0.17	0.24	0.31	0.28
Embedded large S=128	0.08	0.18	0.25	0.26	0.31
Embedded extra large S=320	0.08	0.18	0.24	0.24	0.34

Lifecycle Phase Values of τ_p

Mode & Code Size	Plan & Requirements	System Design	Detailed Design	Module Code & Test	Integration & Test
Organic Small S=2	0.10	0.19	0.24	0.39	0.18
Organic medium S=32	0.12	0.19	0.21	0.34	0.26
Semidetached medium S=32	0.20	0.26	0.21	0.27	0.26
Semidetached large S=128	0.22	0.27	0.19	0.25	0.29
Embedded large S=128	0.36	0.36	0.18	0.18	0.28
Embedded extra large S=320	0.40	0.38	0.16	0.16	0.30